

AN ENERGY BOUND FOR AVOIDANCE INDICATORS, AND FOURIER TAILS OF DNFs OVER PRODUCT SPACES

ANONYMOUS

ABSTRACT. Let F be the indicator that none of a finite family of cylinder events occurs, on an arbitrary finite product probability space, and let $F = \sum_U F^{=U}$ be its Efron–Stein decomposition. We prove that

$$\sum_U \left(\prod_{v \in U} \lambda_v \right) \|F^{=U}\|_2^2 \leq 1$$

for all weights $\lambda_v \geq 1$ such that $\prod_{v \in \text{supp } E} \lambda_v \leq 2$ for every event E . There is no dependence on the number of events, the alphabet sizes or the measure. In particular every width- k DNF $g : \{\pm 1\}^n \rightarrow \{\pm 1\}$ satisfies $\sum_{|S| > t} \hat{g}(S)^2 \leq 4 \cdot 2^{-(t+1)/k}$, under every product measure; so g is ε -concentrated up to degree $k \log_2(4/\varepsilon)$. The proof does not use a switching lemma. It bounds the Efron–Stein energies by signed cover counts of the events that hold at a random point, writes the resulting hypergraph functional as the second moment of an alternating union sum under a random deletion, and bounds that sum by deletion–contraction. Over general product spaces the base $2^{-1/k}$ is sharp, and the threshold 2 is necessary. We also give a version with weights that grow along a filtration of each coordinate, and we explain the application that motivated it: the junta term in a lower bound for the least “multiplier modulus” of the Erdős–Straus equation.

1. INTRODUCTION

Let $\Omega = \prod_{v \in V_0} \Omega_v$ be a finite product of finite sets with a product probability measure $\pi = \bigotimes_v \pi_v$. A *cylinder event* fixes the values of the coordinates in a set $\text{supp } E \subseteq V_0$, its *support*. For a finite family $\mathcal{E}$ of cylinder events let

$$F = F_{\mathcal{E}} = \prod_{E \in \mathcal{E}} (1 - \mathbf{1}_E)$$

be its *avoidance indicator*, and let $F = \sum_{U \subseteq V_0} F^{=U}$ be the Efron–Stein (Hoeffding) decomposition of F in $L^2(\pi)$ [7, 12], [16, §8.3]. For the uniform measure on $\{\pm 1\}^n$ this is the Fourier expansion, $F^{=U} = \hat{F}(U) \chi_U$. We study the weighted energy

$$G_F(\lambda) = \sum_{U \subseteq V_0} \lambda^U \|F^{=U}\|^2, \quad \lambda^U = \prod_{v \in U} \lambda_v, \quad \lambda_v \geq 1.$$

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For constant weights $\lambda_v = \rho^2$ it is $\|T_\rho F\|^2$, where T_ρ multiplies the component $F^{=U}$ by $\rho^{|U|}$; so we are asking for the largest noise operator *with* $\rho > 1$ that still maps every avoidance indicator (of a suitable family) into the unit ball of L^2 . (It is not a contraction in L^2 : $\|T_\rho F\| > \|F\|$ for nonconstant F .)

Theorem 1.1 (Energy bound). *Let $\lambda_v \geq 1$ ($v \in V_0$), and suppose that $\lambda^{\text{supp } E} \leq 2$ for every $E \in \mathcal{E}$. Then*

$$G_F(\lambda) = \sum_U \lambda^U \|F^{=U}\|^2 \leq 1.$$

The hypothesis is on single events only. There is no condition on the number of events, on their masses or overlaps, on the alphabet sizes $|\Omega_v|$ or on the measure. The constant 2 cannot be raised (Proposition 2.3). Since $\sum_{U: \lambda^U > e^\tau} \|F^{=U}\|^2 \leq e^{-\tau} G_F(\lambda)$, the theorem is a family of tail bounds, one for each choice of weights. Constant weights give the following.

Corollary 1.2 (Energy tail). *If every event of $\mathcal{E}$ has $|\text{supp } E| \leq k$ ($k \geq 1$), then for $f \in \{F, 1 - F\}$ and every integer $t \geq 0$,*

$$\text{En}(f; t) := \sum_{|U| > t} \|f^{=U}\|^2 \leq 2^{-(t+1)/k} \mathbb{P}(F = 0)(2 - \mathbb{P}(F = 0)) \leq 2^{-(t+1)/k}.$$

A width- k DNF is $g = 2F - 1$ for the family of its terms (with the convention $-1 = \text{True}$), and the corollary becomes a statement about its Fourier tail.

Corollary 1.3 (Fourier tails of DNFs). *Let $g : \{\pm 1\}^n \rightarrow \{\pm 1\}$ be computed by a DNF of width k , and let π be any product measure on $\{\pm 1\}^n$ (Efron–Stein components in $L^2(\pi)$; Fourier coefficients if π is uniform). Then for every integer $t \geq 0$*

$$W^{>t}[g] := \sum_{|U| > t} \|g^{=U}\|^2 \leq 4p(2-p) \cdot 2^{-(t+1)/k} \leq 4 \cdot 2^{-(t+1)/k},$$

where $p = \mathbb{P}[g = -1]$. So g is ε -concentrated up to degree $k \log_2(4/\varepsilon)$. Moreover $\sum_U 2^{|U|/k} \|g^{=U}\|^2 \leq 1 + 4p$, and the total influence $\mathbf{I}[g] := \sum_U |U| \|g^{=U}\|^2$ is at most $(4k/\ln 2)p$. The same holds for q -ary DNFs, with literals $x_v \in S_v$, under any product measure on $\prod_v [q_v]$.

Comparison. The classical bound goes back to Linial, Mansour and Nisan [14], via Håstad’s switching lemma [9]: width- w DNFs are ε -concentrated up to degree $O(w \log(1/\varepsilon))$. In the form of O’Donnell’s book [16, §4.4] it reads $W^{\geq t}[g] \leq 2 \cdot 2^{-t/(20w)}$ (we quote the constant 20 from memory and could not re-check it). It has been used in this form, with an unspecified constant C , in later work on DNFs, for instance by Mansour [15] (from memory) and by Lecomte and Tan [13, Fact 6]. Corollary 1.3 has the constant 1 in the exponent and holds for arbitrary product measures and alphabets. Over general product spaces the base $2^{-1/k}$ is sharp: no bound $C\rho^{-t/k}$ with $\rho > 2$ holds, and at most a factor of order $\sqrt{t/k}$ can be gained (Proposition 6.1).

On the uniform cube we only know that the true rate lies between $3^{-t/k}$ and $2^{-t/k}$ (Example 6.2). The influence bound is *not* a gain: an elementary certificate argument gives $\mathbf{I}[g] \leq 2kp$ on the uniform cube and $\sum_v \|L_v g\|^2 \leq 4kp$ under any product measure (Remark 5.1), so our $(4k/\ln 2)p$ is weaker by the factor $2/\ln 2$, resp. $1/\ln 2$. Sharper bounds without the factor p are known (from memory: $\mathbf{I} \leq 2w$ is usually attributed to Boppana [5], and we recall that Amano [1] improved it to $\mathbf{I} \leq w$). We state the influence bound only because it falls out.

The nearest prior art we know is the paper of Lecomte and Tan [13]. They bound $|\hat{g}(S)|$ by $2^{|S|}$ times the probability that S is covered by the terms satisfied at a random point. This is the device of our Lemma 3.1, but there are three differences: our cover counts are *signed*, and vanish as soon as a satisfied term misses S ; we work in L^2 with all levels and weights at once; and our degree concentration does not go through the switching lemma. The two combinatorial steps of the proof, a polarization identity (Lemma 4.1) and a deletion-contraction bound (Lemma 4.2), have no counterpart there.

Why the switching-lemma constant is large (heuristic). The following re-derivation uses two inputs quoted from memory: Håstad's bound $\mathbb{P}_\rho[\text{DT}(g|_\rho) \geq d] \leq (Apw)^d$ with $A \approx 5$ for a p -random restriction [9], [16, §4.4], and the LMN reduction $W^{\geq t}[g] \leq 2\mathbb{P}_\rho[\text{DT}(g|_\rho) \geq pt/2]$ [14]. With $d = pt/2$ and $x = Apw$ the bound is $2 \exp(\frac{t}{2Aw} x \ln x)$, which is smallest at $x = 1/e$, giving $W^{\geq t}[g] \leq 2 \cdot 2^{-t/(c_0 w)}$ with $c_0 = 2eA \ln 2 \approx 18.8$. This is consistent with the constant 20 quoted above. The loss is intrinsic to the route: the restriction keeps only a p -fraction of the degree, and the switching base forces $pw \lesssim 1/(eA)$.

What we checked. We have no internet access while writing this draft, and our literature search is partial. Earlier, [13] and Tal [20] were checked online; in [20] we found no explicit-constant tail bound for width- w DNFs. Everything else in this section is quoted from memory and is marked as such. We believe Theorem 1.1 and the constant 1 in Corollary 1.3 are new, but we make no priority claim. Before submission the following must be checked; all descriptions in this list are from memory and may be inaccurate.

- Sharpenings of LMN and of the switching lemma: Håstad's "slight sharpening of LMN" [10]; Razborov's combinatorial proof and Beame's exposition [3]; Rossman's entropy proof [18]; Håstad's work on the correlation of parity with small-depth circuits [11]. As far as we recall these improve the dependence on size and depth for AC^0 , and none states a width- w DNF tail with exponent constant 1.
- Product distributions: Furst, Jackson and Smith [8] extended LMN-type concentration to product distributions on $\{\pm 1\}^n$, with a dependence on the bias that we do not remember; Blais, O'Donnell and Wimmer [4] study concentration and learning over arbitrary finite product domains. Corollary 1.3 has no bias dependence at all. A comparison with these papers is needed.

- Total influence of k -CNFs and DNFs: Boppana [5], Amano [1] (see Remark 5.1).
- Fourier growth and noise operators with $\rho > 1$: G_F at constant weight is $\|T_\rho F\|^2$ with $\rho > 1$. Bounds of Fourier-growth type for AC^0 are in [20] and, we believe, in later work of Chattopadhyay, Hatami, Hosseini and Lovett on pseudorandom generators from Fourier growth [6]. We have not checked whether an inequality like Theorem 1.1 appears in this literature.

Proof outline. Write $G_F = \sum_V \mu^V \|L_V F\|^2$ with $\mu_v = \lambda_v - 1$ and $L_V = \prod_{v \in V} (I - \mathbb{E}_v)$. Section 3 bounds $\|L_V F\|^2$ by the mean square of a signed count $N_{\mathcal{H}(x)}(V)$ of covers of V by the events that hold at x , and so reduces Theorem 1.1 to the hypergraph inequality $Q_\mu(\mathcal{H}) = \sum_V \mu^V N_{\mathcal{H}}(V)^2 \leq 1$. Section 4 proves it: $Q_\mu(\mathcal{H})$ is the second moment of the alternating sum $\Theta_\lambda(\mathcal{H}_P) = \sum_{\mathcal{J} \subseteq \mathcal{H}_P} (-1)^{|\mathcal{J}|} \lambda^{\bigcup \mathcal{J}}$ over the edges that miss a random set P , and deletion-contraction gives $|\Theta| \leq 1$. Nothing in the proof is specific to two-point spaces or to the uniform measure.

Organisation. Section 2 sets up notation. Sections 3 and 4 prove Theorem 1.1. Section 5 proves the corollaries, Section 6 discusses sharpness, failed strengthenings and open problems, and Section 7 gives the filtration variant. Section 8 sketches the application to the Erdős–Straus equation [19]. The exact-arithmetic script `scripts/energy_note_check.py` checks the main inequalities on small cases (Section 9).

2. SETTING AND ELEMENTARY FACTS

Product spaces. Throughout, $\Omega = \prod_{v \in V_0} \Omega_v$ is a finite product of finite sets, V_0 is finite, and $\pi = \bigotimes_v \pi_v$ is a product probability measure. We write $\mathbb{P}$, $\mathbb{E}$, $\langle \cdot, \cdot \rangle$ and $\|\cdot\| = \|\cdot\|_{L^2(\pi)}$. For $W \subseteq V_0$ we write x_W for the restriction of $x \in \Omega$ to W , $\Omega_W = \prod_{v \in W} \Omega_v$ and $\pi_W = \bigotimes_{v \in W} \pi_v$. A *cylinder event* is a set $E = \{x : x_v = \sigma_v (v \in \text{supp } E)\}$ with $\text{supp } E \subseteq V_0$ and $\sigma \in \Omega_{\text{supp } E}$. A family $\mathcal{E}$ of cylinder events is finite, and repetitions are allowed. Its avoidance indicator is $F = \prod_{E \in \mathcal{E}} (1 - \mathbf{1}_E)$, and $h = 1 - F$ is the indicator that some event occurs. For weights $\lambda_v \geq 1$ we put $\mu_v = \lambda_v - 1 \geq 0$, $\lambda^U = \prod_{v \in U} \lambda_v$, $\mu^U = \prod_{v \in U} \mu_v$, and $w_E = \lambda^{\text{supp } E}$ (the *weight* of E).

Efron–Stein decomposition (classical; [7, 12], [16, §8.3]). For $\varphi : \Omega \rightarrow \mathbb{R}$ and $W \subseteq V_0$ let $\mathbb{E}[\varphi \mid x_W]$ be the conditional expectation given the coordinates in W , and

$$\varphi^{=U} = \sum_{W \subseteq U} (-1)^{|U \setminus W|} \mathbb{E}[\varphi \mid x_W].$$

Then $\varphi = \sum_U \varphi^{=U}$, the components are mutually orthogonal, $\varphi^{=U}$ depends only on x_U , and $\mathbb{E}_u \varphi^{=U} = 0$ for $u \in U$, where $\mathbb{E}_u$ averages over the coordinate u (with respect to π_u). Put $L_u = I - \mathbb{E}_u$ and $L_V = \prod_{u \in V} L_u$. The L_u are commuting orthogonal projections, and $L_V \varphi = \sum_{U \supseteq V} \varphi^{=U}$. If $\mathcal{D}$ is a down-closed family of subsets of V_0 , then $\varphi^{\mathcal{D}} = \sum_{U \in \mathcal{D}} \varphi^{=U}$ is a linear combination

of the functions $\mathbb{E}[\varphi \mid x_W]$, $W \in \mathcal{D}$ (by Möbius inversion), and

$$\mathbb{E}(\varphi - \varphi^{\mathcal{D}})^2 = \sum_{U \notin \mathcal{D}} \|\varphi^{\neg U}\|^2. \quad (1)$$

For $\mathcal{D} = \{U : |U| \leq t\}$ this is the *energy above level t* , $\text{En}(\varphi; t)$. It is the least value of $\mathbb{E}(\varphi - g)^2$ over sums g of functions that each depend on at most t coordinates. For ± 1 -valued g we also write $W^{>t}[g] = \text{En}(g; t)$, as in [16].

Definition 2.1 (Energy generating function). For weights $\lambda_v \geq 1$,

$$G_\varphi(\lambda) = \sum_{U \subseteq V_0} \lambda^U \|\varphi^{\neg U}\|^2.$$

Lemma 2.2 (Elementary identities). *For every φ and all weights $\lambda_v \geq 1$:*

- (a) $G_\varphi(\lambda) = \sum_{V \subseteq V_0} \mu^V \|L_V \varphi\|^2$;
- (b) $\sum_{U: \lambda^U > e^\tau} \|\varphi^{\neg U}\|^2 \leq e^{-\tau} G_\varphi(\lambda)$ for every real τ ;
- (c) $G_{1-\varphi} = 1 - 2\mathbb{E}\varphi + G_\varphi$; in particular $G_F \leq 1$ if and only if $G_h \leq 2\mathbb{E}h$;
- (d) if $\varphi(x) = \prod_v \varphi_v(x_v)$ is a product, then $G_\varphi(\lambda) = \prod_v G_{\varphi_v}(\lambda_v)$, where $G_{\varphi_v}(\lambda_v) = (\mathbb{E}\varphi_v)^2 + \lambda_v \text{Var } \varphi_v$.

Proof. (a) Expand $\lambda^U = \prod_{v \in U} (1 + \mu_v) = \sum_{V \subseteq U} \mu^V$ and use $\|L_V \varphi\|^2 = \sum_{U \supseteq V} \|\varphi^{\neg U}\|^2$. (b) is Markov's inequality. (c) $(1 - \varphi)^{\neg U} = -\varphi^{\neg U}$ for $U \neq \emptyset$ and $(1 - \varphi)^{\neg \emptyset} = 1 - \mathbb{E}\varphi$. (d) For a product, $\varphi^{\neg U} = \prod_{v \in U} (\varphi_v - \mathbb{E}\varphi_v) \prod_{v \notin U} \mathbb{E}\varphi_v$. $\square$

Proposition 2.3 (A single event; the threshold 2). *Let E fix $x_v = \sigma_v$ on $\text{supp } E$, put $p_v = \pi_v(\sigma_v)$ and $\pi_E = \mathbb{P}(E) = \prod_{v \in \text{supp } E} p_v$. Then*

$$G_{1-\mathbf{1}_E}(\lambda) = 1 - 2\pi_E + \pi_E \prod_{v \in \text{supp } E} (\lambda_v - \mu_v p_v).$$

Hence, for fixed weights with $w_E > 2$, $G_{1-\mathbf{1}_E}(\lambda) > 1$ as soon as all $p_v > 0$ are small enough. So the constant 2 in Theorem 1.1 cannot be replaced by any larger number, even for a single event.

Proof. By Lemma 2.2(d), with $\varphi_v = \mathbf{1}[x_v = \sigma_v]$, $G_{\mathbf{1}_E} = \prod_v (p_v^2 + \lambda_v p_v (1 - p_v)) = \pi_E \prod_v (\lambda_v - \mu_v p_v)$. Then use Lemma 2.2(c). The product tends to w_E as $p \rightarrow 0$. $\square$

On the uniform cube $\{\pm 1\}^n$ the single-event condition is the weaker $\prod_v (1 + \lambda_v)/2 \leq 2$. The threshold $w_E \leq 2$ is forced by large alphabets or strongly biased measures.

3. THE COVER BOUND

A *hypergraph* $\mathcal{H}$ is a finite set of subsets (*edges*) of V_0 ; the empty edge is allowed. For $V \subseteq V_0$ let

$$N_{\mathcal{H}}(V) = \sum_{\mathcal{J} \subseteq \mathcal{H}: \bigcup_{J \in \mathcal{J}} (J \cap V) = V} (-1)^{|\mathcal{J}|}, \quad Q_{\mu}(\mathcal{H}) = \sum_{V \subseteq V_0} \mu^V N_{\mathcal{H}}(V)^2.$$

So $N_{\mathcal{H}}(V)$ is a signed count of the subfamilies whose traces cover V . It depends only on the multiset of traces $J \cap V$. It does not change if an edge is replaced by several edges with the same trace: a class of $m \geq 1$ edges with equal traces contributes $\sum_{j \geq 1} \binom{m}{j} (-1)^j = -1$ to the choice of whether the class meets $\mathcal{J}$, exactly like a single edge. If some edge misses V , then $N_{\mathcal{H}}(V) = 0$: toggling that edge is a sign-reversing involution on the covering subfamilies. Also $N_{\mathcal{H}}(\emptyset) = \mathbf{1}[\mathcal{H} = \emptyset]$. For example, $Q_{\mu}(\{J\}) = \lambda^J - 1$, and for pairwise disjoint edges $Q_{\mu}(\{J_1, \dots, J_m\}) = \prod_i (\lambda^{J_i} - 1)$.

Lemma 3.1 (Cover bound). *For every family $\mathcal{E}$ of cylinder events and every $V \subseteq V_0$,*

$$\|L_V F\|^2 \leq \mathbb{E}_x[N_{\mathcal{H}(x)}(V)^2], \quad \mathcal{H}(x) = \{\text{supp } E : E \in \mathcal{E}, x \in E\}.$$

Consequently $G_F(\lambda) \leq \mathbb{E}_x[Q_{\mu}(\mathcal{H}(x))]$ for all weights $\lambda_v \geq 1$.

Proof. Write $x = (\sigma, y)$ with $\sigma \in \Omega_V$ and $y \in \Omega_{V^c}$, $V^c = V_0 \setminus V$. The operator L_V acts on the V -coordinates only, so $\|L_V F\|^2 = \mathbb{E}_y \|L_V g_y\|_{L^2(\pi_V)}^2$ with $g_y = F(\cdot, y)$. Now g_y is the avoidance indicator on Ω_V of the *restricted* events: those $E \in \mathcal{E}$ whose conditions off V hold at y , restricted to $\text{supp } E \cap V$. Expand

$$g_y = \prod_E (1 - \mathbf{1}_{E|_y}) = \sum_{\mathcal{J}} (-1)^{|\mathcal{J}|} \mathbf{1}_{\mathcal{J}}$$

over subfamilies $\mathcal{J}$ of restricted events, where $\mathbf{1}_{\mathcal{J}}$ is the indicator of their intersection, which is a cylinder on Ω_V with support $\bigcup_{E \in \mathcal{J}} (\text{supp } E \cap V)$, or it is empty. The projection L_V annihilates every function that does not depend on all coordinates in V , hence every cylinder whose support is a proper subset of V . A cylinder with support V is a point indicator $\mathbf{1}_{\{\sigma\}}$. Therefore $L_V g_y = L_V c_y$, where

$$c_y(\sigma) = \sum_{\mathcal{J}: \text{supp } \mathbf{1}_{\mathcal{J}} = V, \mathbf{1}_{\mathcal{J}} = \mathbf{1}_{\{\sigma\}}} (-1)^{|\mathcal{J}|}.$$

Since L_V is an orthogonal projection in $L^2(\pi_V)$, $\|L_V c_y\|^2 \leq \mathbb{E}_{\sigma} c_y(\sigma)^2$. The subfamilies counted in $c_y(\sigma)$ are exactly the subfamilies of events that hold at $x = (\sigma, y)$ whose traces on V cover V . Two events that hold at x and have the same support are the same cylinder. By the remark on repeated traces, $c_y(\sigma) = N_{\mathcal{H}(x)}(V)$. Taking $\mathbb{E}_y$ gives the first claim. For the second, multiply by μ^V , sum over V and use Lemma 2.2(a). $\square$

The lemma is an equality for $V = \emptyset$ (both sides equal $\mathbb{E}F$). The only loss is the projection step $\|L_V c\| \leq \|c\|$. Two features matter below. First, the suppression is built in: $N_{\mathcal{H}(x)}(V) = 0$ as soon as some event that holds at x misses V . Second, the right-hand side no longer involves the measure, except through the law of $\mathcal{H}(x)$; Theorem 1.1 will follow from a bound on Q_{μ} that is uniform over all hypergraphs whose edges have weight at most 2.

Remark 3.2 (Unsigned counts are too weak). Lecomte and Tan [13, Fact 9] bound $|\hat{g}(S)|$ for a DNF g on the uniform cube by $2^{|S|} \mathbb{P}_x[S \text{ is covered by}]$

the terms satisfied by x]; we quote the form from [13], checked earlier. The indicator “ S is covered” does not see cancellations. For example, if $\mathcal{H}(x)$ contains a singleton $\{u\}$ and an edge containing a set $V \not\ni u$, then $N_{\mathcal{H}(x)}(V) = 0$, while $\mathbf{1}[V \text{ is covered}] = 1$, so the bound of [13] charges $2^{|V|}$ at such points.

4. THE HYPERGRAPH INEQUALITY

For a finite multiset $\mathcal{C}$ of subsets of V_0 (empty members allowed) put

$$\Theta_\lambda(\mathcal{C}) = \sum_{\mathcal{J} \subseteq \mathcal{C}} (-1)^{|\mathcal{J}|} \lambda^{\bigcup \mathcal{J}} \quad (\Theta_\lambda(\emptyset) = 1),$$

the sum running over sub-multisets (members are distinguishable).

Lemma 4.1 (Polarization). *Let $\lambda_v \geq 1$, $\mu_v = \lambda_v - 1$, and let $P \subseteq V_0$ be random with $\mathbb{P}(v \in P) = \mu_v/\lambda_v$, independently for different v . For every hypergraph $\mathcal{H}$,*

$$Q_\mu(\mathcal{H}) = \mathbb{E}_P[\Theta_\lambda(\mathcal{H}_P)^2], \quad \mathcal{H}_P = \{J \in \mathcal{H} : J \cap P = \emptyset\}.$$

Proof. Let $\tau(R) = \mathbf{1}[R \cap J \neq \emptyset \text{ for all } J \in \mathcal{H}]$ for $R \subseteq V_0$, and $\hat{\tau}(V) = \sum_{R \subseteq V} (-1)^{|V \setminus R|} \tau(R)$. By inclusion–exclusion,

$$\tau(R) = \sum_{\mathcal{J} \subseteq \mathcal{H}} (-1)^{|\mathcal{J}|} \mathbf{1}[R \cap \bigcup \mathcal{J} = \emptyset]. \quad (2)$$

The Möbius transform of $R \mapsto \mathbf{1}[R \cap U = \emptyset]$ is $V \mapsto (-1)^{|V|} \mathbf{1}[V \subseteq U]$, so $\hat{\tau}(V) = (-1)^{|V|} N_{\mathcal{H}}(V)$. Let

$$\mathcal{T}(p) = \sum_V \hat{\tau}(V) p^V = \sum_R \tau(R) \prod_{v \in R} p_v \prod_{v \notin R} (1 - p_v)$$

be the multilinear extension of τ . If the p_v are independent random variables with mean 0 and variance μ_v , then distinct monomials are orthogonal and $\mathbb{E}[\mathcal{T}(p)^2] = \sum_V \mu^V \hat{\tau}(V)^2 = Q_\mu(\mathcal{H})$, whatever the laws of the p_v . Take $p_v = 1$ with probability μ_v/λ_v and $p_v = -\mu_v$ otherwise. The mean is $\mu_v/\lambda_v - \mu_v/\lambda_v = 0$ and the variance is $\mu_v/\lambda_v + \mu_v^2/\lambda_v = \mu_v$. Let $P = \{v : p_v = 1\}$. In the second form of $\mathcal{T}$, the factor $1 - p_v$ vanishes for $v \in P$, so only the $R \supseteq P$ contribute; writing $R = P \cup B$ with $B \subseteq P^c$, and using $1 - p_v = \lambda_v$ for $v \notin P$,

$$\mathcal{T}(p) = \sum_{B \subseteq P^c} (-\mu)^B \lambda^{P^c \setminus B} \tau(P \cup B).$$

An edge that meets P meets $P \cup B$, so (2) for the hypergraph $\mathcal{H}_P$ gives $\tau(P \cup B) = \sum_{\mathcal{J} \subseteq \mathcal{H}_P} (-1)^{|\mathcal{J}|} \mathbf{1}[B \cap \bigcup \mathcal{J} = \emptyset]$. For $U \subseteq P^c$,

$$\sum_{B \subseteq P^c \setminus U} (-\mu)^B \lambda^{P^c \setminus B} = \lambda^U \prod_{v \in P^c \setminus U} (\lambda_v - \mu_v) = \lambda^U.$$

Hence $\mathcal{T}(p) = \sum_{\mathcal{J} \subseteq \mathcal{H}_P} (-1)^{|\mathcal{J}|} \lambda^{\bigcup \mathcal{J}} = \Theta_\lambda(\mathcal{H}_P)$. $\square$

Lemma 4.2 (Matching bound). *If every member of $\mathcal{C}$ has $w_J := \lambda^J \leq 2$, then for every matching $\mathcal{M} \subseteq \mathcal{C}$ (pairwise disjoint members)*

$$|\Theta_\lambda(\mathcal{C})| \leq \prod_{J \in \mathcal{M}} (w_J - 1); \quad \text{in particular } |\Theta_\lambda(\mathcal{C})| \leq 1.$$

Proof. For $v \in V_0$ let $\mathcal{C} - v$ be the multiset of members not containing v , and $\mathcal{C}/v = \{J \setminus \{v\} : J \in \mathcal{C}\}$ (as a multiset). Split the sum over $\mathcal{J}$ according to whether $\mathcal{J}$ contains a member through v . If it does, then $\lambda^{\bigcup \mathcal{J}} = \lambda_v \lambda^{\bigcup (\mathcal{J}/v)}$; the terms without such a member form $\Theta(\mathcal{C} - v)$ both in $\Theta(\mathcal{C})$ and in $\Theta(\mathcal{C}/v)$. Hence

$$\Theta(\mathcal{C}) = \Theta(\mathcal{C} - v) + \lambda_v [\Theta(\mathcal{C}/v) - \Theta(\mathcal{C} - v)] = \lambda_v \Theta(\mathcal{C}/v) - \mu_v \Theta(\mathcal{C} - v). \quad (3)$$

Both families again have all weights at most 2, since $w_{J \setminus v} = w_J / \lambda_v$ for $J \ni v$ and the other members are unchanged. We induct on $|\bigcup \mathcal{C}|$. If $\emptyset \in \mathcal{C}$, then $\Theta(\mathcal{C}) = 0$ (toggling the empty member pairs the terms), and the claim holds. If $\mathcal{C} = \emptyset$, then $\Theta = 1$ and $\mathcal{M} = \emptyset$. This covers the base case $\bigcup \mathcal{C} = \emptyset$. If $\mathcal{M} = \emptyset$ and $\mathcal{C} \neq \emptyset$, it suffices to prove the bound for $\mathcal{M} = \{J\}$ with any $J \in \mathcal{C}$, since $w_J - 1 \leq 1$. So let $J_0 \in \mathcal{M}$ be nonempty, $v \in J_0$, and $\Pi' = \prod_{J \in \mathcal{M} \setminus \{J_0\}} (w_J - 1)$. In $\mathcal{C}/v$ the family $(\mathcal{M} \setminus \{J_0\}) \cup \{J_0 \setminus \{v\}\}$ is a matching, so by induction $|\Theta(\mathcal{C}/v)| \leq (w_{J_0}/\lambda_v - 1)\Pi'$. In $\mathcal{C} - v$ the family $\mathcal{M} \setminus \{J_0\}$ is a matching, so $|\Theta(\mathcal{C} - v)| \leq \Pi'$. By (3), $|\Theta(\mathcal{C})| \leq (w_{J_0} - \lambda_v)\Pi' + \mu_v \Pi' = (w_{J_0} - 1)\Pi'$. $\square$

The hypothesis $w_J \leq 2$ enters only through $w_J - 1 \leq 1$ and its stability under contraction. Without it Θ is unbounded: for m disjoint singletons of weight λ , $\Theta = (1 - \lambda)^m$.

Theorem 4.3 (Hypergraph inequality). *If every edge of a hypergraph $\mathcal{H}$ has $w_J = \lambda^J \leq 2$, then*

$$Q_\mu(\mathcal{H}) \leq \prod_{J \in \mathcal{M}} (w_J - 1) \quad \text{for every matching } \mathcal{M} \subseteq \mathcal{H};$$

in particular $Q_\mu(\mathcal{H}) \leq 1$, and $Q_\mu(\mathcal{H}) \leq \min_{J \in \mathcal{H}} w_J - 1$ if $\mathcal{H} \neq \emptyset$. Equality holds if $\mathcal{H} = \mathcal{M}$ is itself a matching.

Proof. By Lemma 4.1, and Lemma 4.2 applied to $\mathcal{H}_P$ with the matching $\mathcal{M} \cap \mathcal{H}_P$, $Q_\mu(\mathcal{H}) \leq \mathbb{E}_P \prod_{J \in \mathcal{M} \cap \mathcal{H}_P} (w_J - 1)^2$. The members of $\mathcal{M}$ are disjoint, so the events $\{J \cap P = \emptyset\}$, $J \in \mathcal{M}$, are independent, with probabilities $\prod_{v \in J} (1 - \mu_v / \lambda_v) = 1/w_J$. Hence

$$Q_\mu(\mathcal{H}) \leq \prod_{J \in \mathcal{M}} \left[\left(1 - \frac{1}{w_J}\right) + \frac{(w_J - 1)^2}{w_J} \right] = \prod_{J \in \mathcal{M}} (w_J - 1).$$

The equality case is the example after the definition of Q_μ . $\square$

Proof of Theorem 1.1. By Lemma 3.1, $G_F(\lambda) \leq \mathbb{E}_x [Q_\mu(\mathcal{H}(x))]$. Every edge of $\mathcal{H}(x)$ is the support of an event, so it has weight at most 2, and Theorem 4.3 gives $Q_\mu(\mathcal{H}(x)) \leq 1$ for every x . $\square$

Theorem 4.3 with the matching bound gives slightly more: $G_F(\lambda) \leq \mathbb{E}_x \min_{\mathcal{M}} \prod_{J \in \mathcal{M}} (w_J - 1)$, the minimum over matchings in $\mathcal{H}(x)$. It would be interesting to know whether the minimum over matchings can be replaced by a fractional-matching version (Conjecture 6.5).

5. ENERGY TAILS AND DNFs

Proof of Corollary 1.2. Take $\lambda_v = \lambda = 2^{1/k}$ for all v ; then $w_E = \lambda^{|\text{supp } E|} \leq 2$, and Theorem 1.1 gives $G_F(\lambda) \leq 1$. Let $a = \mathbb{P}(F = 0) = \mathbb{E}h$. Since $\|F^{\emptyset}\|^2 = (\mathbb{E}F)^2 = (1 - a)^2$,

$$\sum_{U \neq \emptyset} \lambda^{|U|} \|F^{\neg U}\|^2 \leq 1 - (1 - a)^2 = a(2 - a).$$

For $t \geq 0$ every U with $|U| > t$ is nonempty and has $\lambda^{|U|} \geq 2^{(t+1)/k}$, so $\text{En}(F; t) \leq 2^{-(t+1)/k} a(2 - a)$. Finally $(1 - F)^{\neg U} = -F^{\neg U}$ for $U \neq \emptyset$. $\square$

Proof of Corollary 1.3. Let g be computed by the DNF $T_1 \vee \dots \vee T_m$ with $-1 = \text{True}$, and let E_i be the set of inputs that satisfy the term T_i . A term is a conjunction of at most k literals on distinct variables (a term containing x_v and $\bar{x}_v$ is never satisfied and may be dropped), so E_i is a cylinder event with $|\text{supp } E_i| \leq k$. Then $g = 2F - 1$ with $F = \prod_i (1 - \mathbf{1}_{E_i})$, and $p = \mathbb{P}[g = -1] = \mathbb{P}(F = 0)$. As $g^{\neg U} = 2F^{\neg U}$ for $U \neq \emptyset$, the first claim is Corollary 1.2 multiplied by 4, and $p(2 - p) \leq 1$. If $4 \cdot 2^{-(d+1)/k} \leq \varepsilon$, i.e. $d + 1 \geq k \log_2(4/\varepsilon)$, then $W^{>d}[g] \leq \varepsilon$; the least such integer d is at most $k \log_2(4/\varepsilon)$.

For the weighted sum, $\|g^{\emptyset}\|^2 = (1 - 2p)^2$ and, with $\lambda = 2^{1/k}$,

$$\sum_{U \neq \emptyset} \lambda^{|U|} \|g^{\neg U}\|^2 = 4(G_h - p^2),$$

where $G_h = G_F - 1 + 2p \leq 2p$ by Lemma 2.2(c). So the sum is at most $(1 - 2p)^2 + 8p - 4p^2 = 1 + 4p$. For the influence, $\sum_U (\lambda^{|U|} - 1) \|F^{\neg U}\|^2 = G_F - \mathbb{E}F \leq p$, and $\lambda^{|U|} - 1 \geq |U| \ln \lambda = |U| \ln 2/k$. Hence

$$\mathbf{I}[g] = \sum_U |U| \|g^{\neg U}\|^2 = 4 \sum_U |U| \|F^{\neg U}\|^2 \leq (4k/\ln 2) p.$$

For a q -ary DNF, a term $\bigwedge_{v \in S} (x_v \in S_v)$ is the disjoint union of the cylinder events $\{x_S = \sigma\}$, $\sigma \in \prod_{v \in S} S_v$, which all have support S . Replacing each term by these events changes neither F nor the supports. $\square$

For a width- k CNF h , $-h$ is a width- k DNF and $W^{>t}[h] = W^{>t}[-h]$; so the same bounds hold with $p = \mathbb{P}[h = +1]$ (the probability that h is false), and the bound $4 \cdot 2^{-(t+1)/k}$ holds as stated. Here $\mathbf{I}[g] = \sum_U |U| \|g^{\neg U}\|^2 = \sum_v \|L_v g\|^2$ is the Efron–Stein total influence; it is the usual total influence for the uniform measure.

Remark 5.1 (The influence bound is elementary). Let g be a width- k DNF and $p = \mathbb{P}[g = -1]$. If $g(x) = -1$, fix a term satisfied at x ; changing a coordinate outside it leaves $g = -1$. So the fibre of x in direction v is

non-constant for at most k coordinates v . Since $\|L_v g\|^2 = \mathbb{E} \text{Var}_{x_v}(g)$ and $\text{Var}_{x_v}(g) = 4q(1-q) \leq 4q$ on a non-constant fibre, with q the conditional probability of $g = -1$, we get $\sum_v \|L_v g\|^2 \leq 4kp$ under any product measure. On the uniform cube, $\mathbf{I}[g]$ is the mean number of sensitive coordinates; every sensitive edge has exactly one endpoint with $g = -1$, so $\mathbf{I}[g] \leq 2kp$. Corollary 1.3 gives the weaker $(4k/\ln 2)p$.

Remark 5.2 (Non-uniform widths). Theorem 1.1 also handles terms of very different widths, and weights adapted to the variables. For instance, if every term contains at most k_1 variables from a set A and at most k_2 from its complement, then $\lambda_v = 2^{1/(2k_1)}$ on A and $2^{1/(2k_2)}$ off A are admissible. This gives $\sum \|F^=U\|^2 \leq 2^{-\tau}$ over the U with $|U \cap A|/k_1 + |U \setminus A|/k_2 > 2\tau$. The weights used in Section 8 are of this kind, with $\log \lambda_v$ proportional to the logarithm of the size of the alphabet at v .

Remark 5.3 (What is not claimed). Corollary 1.3 concerns concentration on low *degree* only. It says nothing about the number of significant coefficients (Mansour's sparsity bound and Mansour's conjecture [15], [13]), and it does not give the random-restriction (decision-tree depth) form of the switching lemma. It does not apply to depth-3 circuits or to AC^0 ; we do not know whether the method extends to them.

6. SHARPNESS, LIMITS OF THE METHOD, OPEN PROBLEMS

Proposition 6.1 (The base $2^{-1/k}$ is sharp). *Fix $k \geq 1$ and $j \geq 1$, and let $t = jk - 1$. For every $\eta > 0$ there is a family of cylinder events of width k , on a product space with all alphabets of size 2 and a suitable product measure, or on $[q]^n$ with the uniform measure and q large, such that*

$$\text{En}(F; t) \geq \frac{2^{-(t+1)/k}}{e\sqrt{j}} - \eta.$$

Consequently, no bound $\text{En}(F; t) \leq C\rho^{-(t+1)/k}$ with $\rho > 2$ holds for all width- k systems, and the same is true for $W^{>t}[g]$ of width- k DNFs under biased product measures on $\{\pm 1\}^n$.

Proof. Take m events on pairwise disjoint sets $S_1, \dots, S_m$ of k coordinates, each fixing at every coordinate of its support a value of probability p (the value -1 of a p -biased bit, or one value of $[q]$ with $q = 1/p$), and put $\pi = p^k$. The factors $1 - \mathbf{1}_{E_i}$ are independent, so $F^=U = \prod_i (1 - \mathbf{1}_{E_i})^{=U \cap S_i}$. If U is the union of j of the S_i , Lemma 2.2(d) gives $\|F^=U\|^2 = [\pi(1-p)^k]^j (1-\pi)^{2(m-j)}$, and $|U| = jk$. Hence

$$\text{En}(F; jk - 1) \geq \binom{m}{j} [\pi(1-p)^k]^j (1-\pi)^{2(m-j)} \rightarrow e^{-2s} \frac{s^j}{j!}$$

as $p \rightarrow 0$ with $m = \lfloor s/\pi \rfloor$. With $s = j/2$ and $j! \leq e j^{j+1/2} e^{-j}$ the limit is at least $2^{-j}/(e\sqrt{j})$, and $2^{-j} = 2^{-(t+1)/k}$. $\square$

So at most a factor of order $\sqrt{t/k}$ could be gained in Corollary 1.2 over general product spaces. In the limit $p \rightarrow 0$, the supremum over s of $e^{-2s} \sum_{i \geq j} s^i / i!$ (the limiting value of $\text{En}(F; jk-1)$) behaves like $2 \cdot 2^{-j} / \sqrt{2\pi j}$ as $j \rightarrow \infty$ (heuristically: the term $i = j$ at $s \approx j/2$, and a tail of ratio about $1/2$). We do not prove this asymptotic; numerically the normalised supremum is 0.2255 at $j = 10$ against $2/\sqrt{20\pi} = 0.2523$, and 0.0876 against 0.0892 at $j = 80$ (internal computations). Together with Proposition 2.3 this shows that both the threshold 2 in Theorem 1.1 and the base of Corollary 1.2 are optimal at the level of generality of the theorem. Note that the extremal measures have atoms of probability $p \rightarrow 0$, depending on t : the base is sharp for the supremum over *all* product measures. For a fixed measure whose atoms are bounded below (for instance $[q]^n$ uniform with q fixed, or the uniform cube) the proposition says nothing, and the optimal rate is open (cf. Example 6.2).

Example 6.2 (The uniform cube). On $\{\pm 1\}^n$ with the uniform measure the extremal families above are not available ($p = 1/2$), and the truth may be better. Let g be the OR of s parities on pairwise disjoint blocks $B_1, \dots, B_s$ of k variables. Each parity is a width- k DNF (with 2^{k-1} terms), so g is a width- k DNF. If the i -th parity is “true” when $\chi_{B_i} = -1$, then $F = \prod_i (1 + \chi_{B_i})/2$, and $\text{En}(F; jk-1) = 4^{-s} \sum_{i \geq j} \binom{s}{i}$. Taking s near $4j/3$ and using Stirling’s formula, $\max_s 4^{-s} \binom{s}{j} = 3^{-j+o(j)}$. So the exponential rate of the best width- k bound on the uniform cube lies between $3^{-t/k}$ and $2^{-t/k}$. We do not know where. In random and hill-climbing searches over small Boolean systems (earlier internal computations) we found nothing better than this example.

No one-event-at-a-time proof. It is natural to try to prove Theorem 1.1 by adding the events one at a time. The following example shows that this cannot work in the obvious form. Adding a further event of weight at most 2 to a system can *increase* G_F .

Example 6.3 (Adding an event can increase G_F). Let $n = 2$, $\Omega = [8]^2$ with the uniform measure, $\lambda_1 = \lambda_2 = \sqrt{2}$, and let $\mathcal{E}$ consist of the 49 point events $\{x = (a, b)\}$ with $a \neq 0 \neq b$. Let $A = \{x = (0, 0)\}$. All events have weight 2, and $G_{F_{\mathcal{E}}} < G_{F_{\mathcal{E} \cup \{A\}}}$; numerically the values are 0.31571 and 0.31757. (The exact values are computed in Section 9.) A similar example exists on $[5]^3$ with $\lambda_v = 2^{1/3}$: take the 48 point events with exactly two nonzero coordinates and add $A = \{x = (0, 0, 0)\}$; then G_F increases from 0.75406 to 0.75467 (checked by the script of Section 9, with 50-digit arithmetic).

Also the inequality fails beyond the threshold. For n uniform bits, the OR of the n literals $x_v = -1$ and $\lambda_v = 4$ give $G_F = (5/4)^n$ by Lemma 2.2(d) and Proposition 2.3. We state two strengthenings that are supported by numerical evidence but are open.

Conjecture 6.4 (Excess weights). *For every family of cylinder events and all weights $\lambda_v \geq 1$,*

$$G_F(\lambda) \leq \prod_{E \in \mathcal{E}} (1 + \mathbb{P}(E)(w_E - 2)_+).$$

Conjecture 6.5 (Fractional matchings). *If every edge of $\mathcal{H}$ has $w_J \leq 2$, then $Q_\mu(\mathcal{H}) \leq \prod_{J \in \mathcal{H}} (w_J - 1)^{y_J}$ for every fractional matching y ($y_J \geq 0$, $\sum_{J \ni v} y_J \leq 1$ for all v).*

Both were tested numerically on small random systems (several thousand cases on $[q]^n$ with $q \leq 5$, $n \leq 9$; floating-point internal computations, with a numerical linear program over y for Conjecture 6.5; not part of this note), without a violation. For Conjecture 6.4 only the weaker form $G_F \leq \exp(\sum_E \mathbb{P}(E)(w_E - 2)_+)$ was tested. For Conjecture 6.5 equality (up to rounding) is frequent. Example 6.3 shows that a proof of Conjecture 6.4 by induction on the number of events would need a stronger induction hypothesis.

Open problems. (1) Determine the optimal rate for width- k DNFs on the uniform cube (Example 6.2). (2) Is there an analogue of Theorem 1.1 for depth-3 circuits, or for AC^0 , with a weight condition on the bottom layer? (3) Does the signed cover count give sparsity bounds in the direction of Mansour’s conjecture?

7. WEIGHTS ALONG A FILTRATION

In Theorem 1.1 the cost of a component $F^{=U}$ is a product over the coordinates in U . In the application the natural cost of a function of a residue $x_\ell \bmod \ell^f$ depends on *how many* base- ℓ digits it reads, and an event that fixes $x_\ell \bmod \ell^v$ fixes the first v digits. We now prove the corresponding variant.

Setting. Let $\mathcal{P}$ be a finite index set. Each coordinate X_ℓ ($\ell \in \mathcal{P}$) is a tuple of independent *digits* $(\delta_{\ell,i})_{i_0(\ell) \leq i < f_\ell}$ with arbitrary distributions on finite sets, and the coordinates are independent. The Efron–Stein decomposition $F = \sum_U F^{=U}$ is taken over the digits, so U is a set of pairs (ℓ, i) ; we write $U_\ell = \{i : (\ell, i) \in U\}$. An event E is a *digit-prefix event* if, for some integers $0 \leq v_\ell(E) \leq f_\ell$, it fixes exactly the digits $\delta_{\ell,i}$ with $i_0(\ell) \leq i < v_\ell(E)$, for each ℓ , to arbitrary values (no digit of X_ℓ if $v_\ell(E) \leq i_0(\ell)$). The offsets are integers $i_0(\ell) \geq 0$. Such an event is a cylinder event on the digits. (The offset $i_0(\ell)$ allows digits below $i_0(\ell)$ to be fixed in advance; they are not coordinates.)

Theorem 7.1 (Filtration energy bound). *Let $\lambda_\ell \geq 1$ ($\ell \in \mathcal{P}$), and suppose that every event of $\mathcal{E}$ is a digit-prefix event with $\prod_\ell \lambda_\ell^{2v_\ell(E)} \leq 2$. Then*

$$\sum_U \left(\prod_{\ell: U_\ell \neq \emptyset} \lambda_\ell^{1 + \max U_\ell} \right) \|F^{=U}\|^2 \leq 1.$$

Applied directly, Theorem 1.1 with digit weights would charge a set U for each digit in it, and the event weights would grow with the number of fixed digits. Here the charge for U_ℓ depends only on its *deepest* digit, which is what a function of X_ℓ that reads U_ℓ costs, and the hypothesis loses only the factor 2 in the exponent.

Proof. Write $\Lambda_{\ell,i} = \lambda_\ell^{i+1}$ for $i \geq i_0(\ell)$, $\Lambda_{\ell,i_0(\ell)-1} = 1$, and $\mu'_{\ell,j} = \Lambda_{\ell,j} - \Lambda_{\ell,j-1} \geq 0$ for $i_0(\ell) \leq j < f_\ell$. For such j let $P_{\ell,j} = I - \mathbb{E}_{B_{\ell,j}}$, where $\mathbb{E}_{B_{\ell,j}}$ averages over the *tail block* $B_{\ell,j} = (\delta_{\ell,i})_{i \geq j}$. Averaging over a digit in U kills $F^{=U}$, and $F^{=U}$ does not depend on the digits outside U . Hence $P_{\ell,j}F^{=U} = F^{=U}$ if $U_\ell \neq \emptyset$ and $\max U_\ell \geq j$, and $P_{\ell,j}F^{=U} = 0$ otherwise. Call a set V of pairs (ℓ, j) a *selection* if it contains at most one pair for each ℓ , and put $L'_V = \prod_{(\ell,j) \in V} P_{\ell,j}$. Then $\|L'_V F\|^2 = \sum_{U \succ V} \|F^{=U}\|^2$, where $U \succ V$ means $\max U_\ell \geq j$ for all $(\ell, j) \in V$. For fixed U , the sum of μ'^V over the selections V with $U \succ V$ factorises and telescopes:

$$\prod_{\ell} \left(1 + \sum_{i_0(\ell) \leq j \leq \max U_\ell} \mu'_{\ell,j}\right) = \prod_{\ell: U_\ell \neq \emptyset} \Lambda_{\ell, \max U_\ell}.$$

So the left side of the claim equals $\sum_V \mu'^V \|L'_V F\|^2$, the sum over selections.

Fix a selection V and pass to the coarser product space whose coordinates are, for $(\ell, j) \in V$, the head $(\delta_{\ell,i})_{i < j}$ and the tail block $B_{\ell,j}$, and, for the other ℓ , the whole X_ℓ . There L'_V is the operator $L_{V'}$ of Section 2 for the set V' of tail blocks. Each event E is a disjoint union of cylinder events on the coarse coordinates with the same coarse support: enumerate the values of the unfixed digits of each coarse coordinate that E involves. Splitting every event in this way does not change F , and exactly one piece of E holds at x if E does. The trace of the coarse support of E on V' is the set of the $B_{\ell,j}$ with $(\ell, j) \in V$ and $j < v_\ell(E)$. Identify $B_{\ell,j}$ with the pair (ℓ, j) and put $\hat{E} = \{(\ell, j) : i_0(\ell) \leq j < v_\ell(E)\}$; then this trace is $\hat{E} \cap V$. Since N depends only on traces, Lemma 3.1 gives

$$\|L'_V F\|^2 \leq \mathbb{E}_x [N_{\hat{\mathcal{H}}(x)}(V)^2], \quad \hat{\mathcal{H}}(x) = \{\hat{E} : E \in \mathcal{E}, x \in E\}.$$

Sum over selections and add the nonnegative terms of all other sets V of pairs: the left side of the claim is at most $\mathbb{E}_x [Q_{\mu'}(\hat{\mathcal{H}}(x))]$. By Theorem 4.3 it remains to show that $\prod_{(\ell,j) \in \hat{E}} (1 + \mu'_{\ell,j}) \leq 2$ for every event. Fix ℓ , write $\lambda = \lambda_\ell$, $v = v_\ell(E)$, $i_0 = i_0(\ell)$, and assume $v > i_0$. Here $1 + \mu'_{\ell,i_0} = \lambda^{i_0+1}$, and $1 + \mu'_{\ell,j} = 1 + \lambda^j(\lambda - 1)$ for $j > i_0$ (if $v \leq i_0$ the factor of ℓ is 1). The hypothesis gives $\lambda^{2v} \leq 2$, so $\lambda^j \leq \lambda^v \leq \sqrt{2} < \lambda + 1$ for $j < v$, which is equivalent to $1 + \lambda^j(\lambda - 1) \leq \lambda^2$. The factor of ℓ is therefore at most $\lambda^{i_0+1} \lambda^{2(v-i_0)} \leq \lambda^{2v}$, and the product over ℓ is at most $\prod_{\ell} \lambda_\ell^{2v_\ell(E)} \leq 2$. $\square$

Corollary 7.2 (Modulus-weighted tail). *In Theorem 7.1 let the $\ell \in \mathcal{P}$ be primes, let $T > 1$, $\mathcal{L} = \log T$, and suppose that $\prod_{\ell} \ell^{v_\ell(E)} \leq T$ for every event. For a set U of digits put $m_U = \prod_{\ell: U_\ell \neq \emptyset} \ell^{1+\max U_\ell}$. Then for every*

$\tau \geq 0$

$$\sum_{U: m_U > e^\tau} \|F^{=U}\|^2 \leq 2^{-\tau/(2\mathcal{L})}.$$

Proof. Take $\lambda_\ell = 2^{\log \ell/(2\mathcal{L})}$. Then $\prod_\ell \lambda_\ell^{2v_\ell(E)} = 2^{\log \prod_\ell \ell^{v_\ell(E)}/\mathcal{L}} \leq 2$, and the weight of U in Theorem 7.1 is $2^{\log m_U/(2\mathcal{L})}$; apply Lemma 2.2(b). $\square$

Remark 7.3 (The factor 2 in the exponent). The proof gives more. If $v > i_0$, the factor of ℓ is

$$\lambda^{i_0+1} \prod_{i_0 < j < v} (1 + \lambda^j(\lambda - 1)) \leq \exp\left(\lambda^{i_0+1} - 1 + \sum_{i_0 < j < v} \lambda^j(\lambda - 1)\right) = \exp(\lambda^v - 1).$$

So it suffices that $\sum_\ell (\lambda_\ell^{v_\ell(E)} - 1) \leq \ln 2$ for every event. In Corollary 7.2 take $\lambda_\ell = 2^{\log \ell/(\rho\mathcal{L})}$ and put $x_\ell = v_\ell(E) \log \ell/\mathcal{L}$, so that $\sum_\ell x_\ell \leq 1$. The function $x \mapsto 2^{x/\rho} - 1$ is convex and vanishes at 0, so it is superadditive, and $\sum_\ell (2^{x_\ell/\rho} - 1) \leq 2^{1/\rho} - 1$. This is at most $\ln 2$ when $\rho \geq \rho_0 := 1/\log_2(1 + \ln 2) = 1.3163\dots$, and then the tail in Corollary 7.2 is at most $2^{-\tau/(\rho\mathcal{L})}$.

8. APPLICATION: ERDŐS–STRAUS WITNESS MODULI

This section explains where Theorem 7.1 is used; the details are in [19], and nothing here is needed for the rest of the note. The Erdős–Straus conjecture asserts that $4/n = 1/x + 1/y + 1/z$ has a solution in positive integers for every $n \geq 2$; it is open. For $M \equiv 3 \pmod{4}$ the classical multiplier identity gives a solution for every n with $n \equiv -uv^{-1} \pmod{M}$ for some factorisation $uvw = (M+1)/4$. Let $W(n)$ be the least such M ($+\infty$ if there is none). In [19] it is shown, assuming Gallagher’s prime number theorem for the family of all characters and four estimates of Elsholtz and Tao on divisor sums, that $W(p) \geq \exp(c(\log p)^{1/5}(\log \log p)^{-1/5})$ for infinitely many primes p . This is an Ω -result about one particular family of explicit solutions, and says nothing about whether the conjecture holds.

The structure of the argument is as follows. Fix T and $\mathcal{L} = \log T$. After a *quarantine* $n \equiv 1 \pmod{Q}$, the condition $W(n) > T$ holds as soon as n avoids a finite family of congruence events $n \equiv b_i \pmod{r_i}$ with $r_i \leq T$, at most T^2 of them. For n distributed by Haar measure on the unit group $\widehat{\mathbb{Z}}^\times$, conditioned on the quarantine, the residues $X_\ell = n \bmod \ell^{f_\ell}$ at the primes involved are independent. Each X_ℓ is uniform on the units modulo ℓ^{f_ℓ} , or on a fibre $1 + \ell^a \mathbb{Z}$ modulo ℓ^{f_ℓ} if ℓ is quarantined with $a = a_\ell \geq 1$. On these sets the uniform measure is the product of the uniform distributions of the free base- ℓ digits (digit 0 ranging over the nonzero residues in the unit case), so the digits are independent coordinates. Moreover an event $n \equiv b \pmod{r}$ is a digit-prefix event with $\prod_\ell \ell^{v_\ell(E)} = r \leq T$. So the avoidance indicator F is in the setting of Corollary 7.2, and so is each conditioned function $F^{(j)}$ (the indicator that no earlier event occurs, restricted to the cylinder of the j -th event): there the digit offsets $i_0(\ell)$ are raised.

To find a *prime* p with $W(p) > T$ one needs a minorant $B \leq F$ that is a combination of congruence classes of small modulus, has mean comparable to $\mathbb{E}F$, and has small ℓ^1 norm. [19] uses the one-sided ℓ^2 sandwich of Bazzi [2] in the form of Razborov [17]:

$$B = 1 - \sum_i \mathbf{1}_{E_i} \left(1 - \sum_{j < i} \mathbf{1}_{E_j} u_j \right)^2 \leq F,$$

$$\mathbb{E}[F - B] \leq m^2 \sum_j \mathbb{P}(E_j) \mathbb{E}[(F^{(j)} - u_j)^2],$$

for any functions u_j of the coordinates not fixed by E_j . The natural choice of u_j is the truncation of the Efron–Stein decomposition of $F^{(j)}$ to digit sets U with $m_U \leq e^\tau$. By (1) it is a combination of classes of modulus at most e^τ (so $\mathbf{1}_{E_j} u_j$ has modulus at most $T e^\tau$, and every class in the expansion of B has modulus at most $T^3 e^{2\tau}$), and by Corollary 7.2 its error is at most $2^{-\tau/(2\mathcal{L})}$. The sandwich needs the error to be about $e^{-O(S)} m^{-2}$, where S is the total mass of the events. So $\tau \ll \mathcal{L}(S + \mathcal{L})$ suffices; in [19] $S \ll \mathcal{L}^4$ (using the Elsholtz–Tao input), and the cell moduli d_i of B satisfy $\log d_i \leq 3\mathcal{L} + 2\tau \ll \mathcal{L}^5$. The quarantine costs $\log Q \ll \mathcal{L}^5 \log \mathcal{L}$, and a linear transfer theorem based on Gallagher’s theorem then produces a prime p with $\log p \ll \mathcal{L}^5 \log \mathcal{L}$.

If instead one encodes the residues in binary and bounds the Fourier tail of $F^{(j)}$ by the switching-lemma route of [14], the junta moduli pick up factors for the width of the events (the number of primes involved) and for the bit length of the residues, since a junta on d bits may read the top bit of each residue. Theorem 7.1 avoids both factors. The exponent $1/5$ needs two further ingredients of [19] (a graded quarantine and a moment bound), and we do not attribute it to Theorem 7.1 alone. Two features of the energy bound matter here: the events are congruences modulo integers up to T , so the alphabets are huge and the measure is far from the Boolean setting, and the cost of a junta must be measured by its modulus rather than by the number of primes it involves.

9. COMPUTATIONAL CHECKS

The proofs above are complete. The script `scripts/energy_note_check.py` is a sanity check of the statements on small cases. It uses exact rational arithmetic throughout; where a statement involves an irrational number, the comparison is rewritten as an exact rational one (for instance $W^{>t} \leq c 2^{-(t+1)/k}$ as $(W^{>t})^k 2^{t+1} \leq c^k$). With seed 1 it checks:

- (A) Theorem 1.1, Lemma 2.2(a) and Lemma 3.1 (for every V) on 150 random systems on $\prod_v [q_v]$ ($n \leq 4$ coordinates, $q_v \leq 5$), with random rational product measures and random rational weights in $[1, 2]$;
- (B) the identity of Lemma 4.1, the two formulas for $N_{\mathcal{H}}$, and the matching bounds of Lemma 4.2 and Theorem 4.3 (including the equality case) on 300 random weighted hypergraphs on at most 5 vertices;

- (C) Corollary 1.3 for *every* Boolean function on 4 bits under the uniform measure, with k its minimal DNF width: the tail bound for all t (exactly), the bounds $1 + 4p$ and $(4k/\ln 2)p$ at the endpoint $\lambda = 2^{1/k}$ (exactly for $k = 1$, in floating point with slack 10^{-12} otherwise), and $G_F(\lambda) \leq 1$ at a rational $\lambda \leq 2^{1/k}$; the largest value of $W^{>t}[g] 2^{(t+1)/k}/(4p(2-p))$ is 0.680;
- (D) the same statements on 150 random DNFs under biased product measures on $\{\pm 1\}^n$ and q -ary DNFs with set-valued literals;
- (E) Theorem 7.1 on 150 random digit-prefix systems in bases 2 and 3, with random offsets $i_0(\ell) \in \{0, 1\}$ and random digit distributions;
- (F) the level polynomial of Proposition 6.1 against brute force, Corollary 1.2 along that family, the formula of Example 6.2, and the sign in both instances of Example 6.3 (exact on $[8]^2$, 50-digit on $[5]^3$). The values $(\max_s \text{En}(F; jk - 1))^{1/j}$ of Example 6.2 (0.299, 0.311, 0.319 for $j = 10, 20, 40$) are printed, not checked.

The run takes about six minutes on one core; its output is in the file `data/energy_note/check_seed1.txt`.

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